

Eutric Planosols:

Eutric Planosols are tropical soils with a light-colored, bleached, sandy or loamy surface horizon that periodically becomes waterlogged, abruptly overlying a dense, clayey subsoil. They are named “Eutric” due to relatively high base saturation (>35%), which makes them more fertile and less acidic than Ferralic Planosols. The topsoil is coarse-textured and low in organic matter, while the subsoil has dense clay with slow permeability. These soils typically occur on flat or gently undulating terrain in seasonally wet areas.

Drainage:

- **Wet season:** Surface horizon may stagnate due to the dense subsoil; overall drainage is better than Ferralic Planosols.
- **Dry season:** Surface horizon dries and becomes firm but remains workable; subsoil slowly permeable.
- **Landscape:** Flat to gently undulating terrain; low-lying areas or basin margins.

Depth:

- Surface horizon: 15–30 cm, sandy or loamy.
- Subsoil: Dense clay layer (>50 cm) with abrupt textural change; roots may penetrate but slower due to dense clay.

Workability:

- **When wet:** Topsoil sticky but manageable; easier to handle than Ferralic Planosols.
- **When dry:** Surface firm and friable; subsoil dense; tillage requires care but is easier than in Ferralic Planosols.

Nutrient Richness & Cation Exchange Capacity (CEC):

- Moderate CEC (15–25 cmol(+)/kg) within 100 cm.
- Base saturation >35%, making them relatively fertile.
- Low phosphorus availability due to fixation; nitrogen and potassium moderate.
- Iron oxides dominate subsoil; organic matter low unless amended.

pH:

- Slightly acidic to near neutral (5.0–6.5).

Management:

- Improve drainage via raised beds, ridges, or contouring to prevent waterlogging.
- Incorporate organic matter (manure, compost) to improve fertility, structure, and water retention.
- Apply phosphorus fertilizers carefully: slow-release rock phosphate with occasional superphosphate.
- Avoid heavy machinery during wet periods to prevent compaction.
- Suitable for crops like wetland rice, fodder crops, and extensive grazing; fertility allows wider crop options than Ferralic Planosols.